Power equations

1

- a $x=3$
- b $x=8$
- c $x=\frac{8}{5}$
- d $x=6$
- e $x=2$
- f $x=6$
- g $x=5$
- h $x=2$

Exponential equations

2

- a $x=8$
- b $x=\frac{2}{9}$
- c $x=3$
- d $x=0$
- e $x=-2$
- f $y=-3$
- g $x=-3$
- h $x=\frac{1}{3}$
- i $n=\frac{1}{3}$
- j $x=-2$

3

- a
  - i $2^{3x}=2^2$
  - ii $x=\frac{2}{3}$
- b
  - i $2^{4x}=2^{-1}$
  - ii $x=-\frac{1}{4}$
- c
  - i $2^{-10}=2^{2x}$
  - ii $x=-5$
- d
  - i $2^{\frac{x}{2}}=2$
  - ii $x=2$

4

- a $y=4$
- b $k=-2$
- c $y=\frac{3}{2}$
- d $x=2$
- e $x=2$
- f $x=\frac{7}{3}$
- g $x=-7$
- h $x=\frac{21}{19}$
- i $y=\frac{11}{6}$
- j $x=-\frac{41}{6}$
- k $x=5$
- l $x=1$
- m $x=1+\frac{2k}{3}$
- n $x=5$
- o $x=\frac{5}{2}$
- p $x=4, x=-1$
- q $x=3$
- r $x=\frac{4}{1+n}$

5

- a
  - i $m=2^x$
  - ii $x=3,\ 0$
- b
  - i $m=2^x$
  - ii $x=3,\ 2$
- c
  - i $m=2^x$
  - ii $x=-1,\ 3$
- d
  - i $m=4^x$
  - ii $x=0,\ 3$
- e
  - i $m=2^x$
  - ii $x=0,\ 2$
- f
  - i $m=3^x$
  - ii $x=1,\ 2$

Exponential equations and logarithms

6

a $3 < x < 4$

b $3 < x < 4$

c $-4 < x < -3$

d There is no real solution.

7

a

i $x = \dfrac{\log 5}{\log 13}$ ii 0.627

b

i $x = \dfrac{\log \frac{1}{11}}{\log 5}$ ii -1.490

c

i $x = \dfrac{\log 2}{\log 3}$ ii 0.631

d

i $x = \dfrac{\log 6.4}{\log 4}$ ii 1.339

e

i $x = \dfrac{\log 5}{\log 0.4}$ ii -1.756

f

i $x = \dfrac{\log 3125}{\log 5}$ ii 5

g

i $x = \dfrac{\log 6}{\log \frac{1}{2}}$ ii -2.585

h

i $x = \dfrac{\log 19\,207}{\log 27}$ ii 2.993

8

a $x = \dfrac{\frac{\log 70}{\log 4} + 8}{2}$

b 5.532

9

a

i Step 2, when bringing the a out of the index it should multiply the logarithm and not add to it.

ii $a = \dfrac{\log 40}{\log 9}$

iii 1.679

b

i Step 3. The 2 and 89 are in the incorrect positions in the logarithm.

ii $a = \dfrac{\log 89}{\log 2}$

iii 6.476

Applications

10
a $t = 5$ hours
b $t = 10$ hours
c $t = 11.55$ hours

11
a 500 mice
b 20%
c 6.03 months

12
a 800 wallabies
b 15%
c 8.53 years

13 6.180 hours